

Predicting Customer Churn for Targeted Retention in the Telecommunications Industry

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Introduction

The telecommunications industry faces intense competition and rising customer acquisition costs. Retaining existing customers is far more profitable than acquiring new ones. While major providers invest heavily in infrastructure, many still rely on reactive churn management and only addressing customers after they cancel.

This project utilizes a predictive machine learning model to identify customers at high risk of churning *before* they leave, enabling the marketing and retention teams to intervene proactively with targeted offers.

Business Problem and Hypothesis

Telecom companies lose millions annually to customer churn. Traditional approaches are reactive and inefficient, often resulting in blanket retention campaigns that waste budget on low-risk or low-value customers.

We hypothesize that customer churn is predictable from a combination of behavioral, contractual, and service-usage patterns. By integrating historical data with engineered risk features (such as Value_at_Risk), our model can identify “latent risk”, customers likely to churn soon, and prioritize high-value accounts for retention efforts. This shifts the company from reactive damage control to proactive, ROI-focused customer retention.

Methods and Analysis

Our analysis utilized the IBM Telco Customer Churn dataset (7,043 records), which was cleaned and standardized for modeling (see Appendix A). The final model-ready dataset

(Telco_Cleaned_Model_Ready.csv) includes demographic, service, billing, and engineered features such as Service_Intensity, Contract_Rank, and Value_at_Risk.

We trained two collaborative models:

- Random Forest Classifier (baseline with class-weight balancing)
- XGBoost Classifier (with scale_pos_weight for imbalance handling)

Hyperparameter tuning was performed with XGBoost using GridSearchCV, optimizing the F1-score for the churn class. Threshold tuning via the precision-recall curve identified the optimal probability cutoff (0.5155) for maximum business utility. Feature importance analysis from the tuned XGBoost model provided clear interpretability of churn drivers (see Figure 1).

Figure 1: Top 15 Features Driving Customer Churn (Tuned XGBoost)

Model: XGBoost was selected as the final model because it delivered superior performance on the minority class (churn) while maintaining strong overall ranking performance (AUC-ROC).

Results: See Figure 1

The tuned XGBoost model achieved strong predictive performance:

- AUC-ROC: **0.844**
(improved from
baseline 0.823)

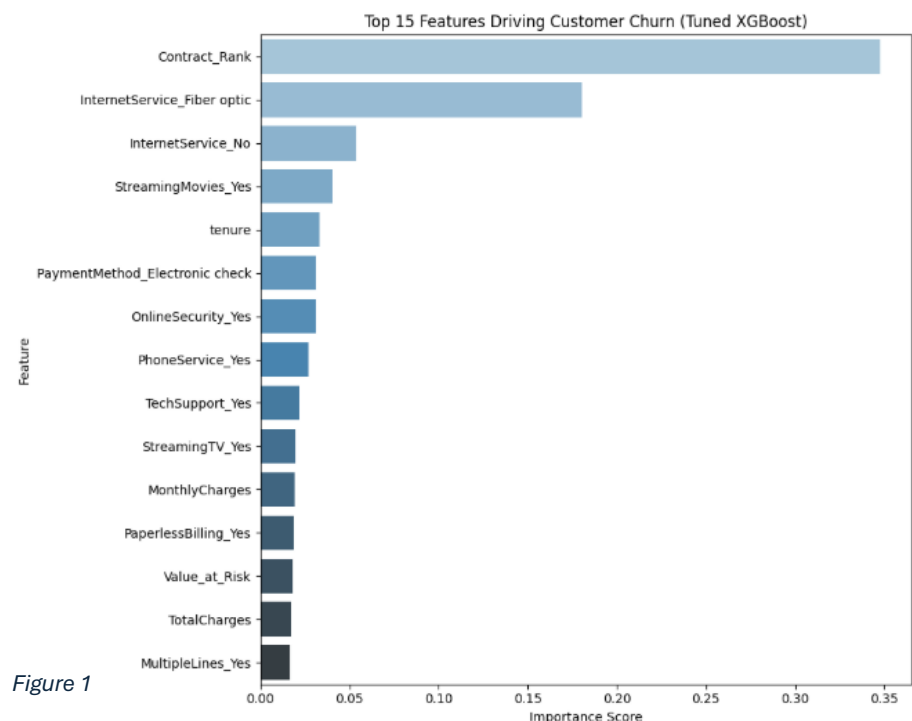


Figure 1

- F1-score (churn class): **0.634** (improved from baseline 0.604)
- Recall on churn class: 0.79

The model flagged **2,718 high-risk customers** using the optimized threshold. These customers were merged with the Customer_Retention_Key.csv and sorted by churn probability and Value_at_Risk to maximize ROI.

Figure 2 Final Model Performance Comparison (Baseline vs. Tuned XGBoost)

	Model	Accuracy	AUC-ROC	F1-Score (Churn Class)
0	Random Forest	0.7921	0.8265	0.5554
1	XGBoost (Baseline)	0.7658	0.8227	0.6043

Figure 2

Key Business Insights:

- The strongest predictors are **Month-to-month contracts**, low **tenure**, high **MonthlyCharges**, **Fiber Optic** internet service, and lack of **Tech Support/Online Security**.
- Churn risk increases significantly when **MonthlyCharges exceed ~\$70–\$80**.
- Highest-risk bundle: Month-to-month + Fiber Optic + high monthly bill + no protective services.

These insights allow the company to focus retention efforts on the most vulnerable and valuable segments.

Figure 3 High-Risk Customer List

Number of high-risk customers (using threshold 0.5155): 2718					
	customerID	Churn	Value_at_Risk	Churn_Probability	Predicted_Churn
4800	9300-AGZNL	1	94.00	0.975548	1
2208	7216-EWTRS	1	100.80	0.975424	1
3380	5178-LMXOP	1	95.10	0.974705	1
1976	9497-QCMMS	1	93.55	0.974160	1
3209	8149-RSOUN	1	93.85	0.971608	1
2577	4910-GMJOT	1	94.60	0.970522	1
6089	8775-LHDJH	1	90.60	0.969722	1
585	5192-EBGOV	1	85.70	0.969188	1
6866	0295-PPHDO	1	95.45	0.968259	1
6482	5419-JPRRN	1	101.45	0.968208	1

Figure 2

Recommendations and Ethical Considerations

Recommendations: To maximize the impact of this model:

- Dispatch retention offers immediately to the top 10–20% of the high-risk list (sorted by churn probability + Value_at_Risk).
- Prioritize month-to-month Fiber Optic customers with monthly bills above \$70 who lack Tech Support or Online Security.
- Pilot a targeted campaign on the highest-Value_at_Risk segment and measure actual retention lift.
- Retrain the model quarterly with new data to maintain accuracy.

Ethical Considerations

- **Impact:** Proactive retention can reduce customer frustration and preserve revenue.

- **Assumptions and Bias:** The model assumes current behavior patterns will continue; sudden external events (e.g., competitor promotions) may affect predictions.
- **Privacy:** All data used is anonymized and compliant with privacy regulations.
- **Misinterpretation:** Risk scores are probabilities, not certainties; they support, rather than replace, human judgment in retention decisions.
- **Responsibility:** Marketing teams should monitor equitable distribution of offers across customer segments.

Conclusion

This project demonstrates that predictive modeling can transform reactive churn management into a proactive, data-driven retention strategy. By combining strong model performance (AUC-ROC 0.844) with clear interpretability and a prioritized high-value customer list, the company can allocate retention resources more efficiently, reduce unnecessary discounts, and protect revenue.

The transition from blanket campaigns to targeted interventions represents a significant step toward sustainable customer loyalty and improved profitability.

References

- IBM. (2019). *Telco Customer Churn Dataset*. Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

Appendix A: Data Preparation and Cleaning

The Telco dataset required several preprocessing steps:

- Converted TotalCharges from string to numeric (handling blanks with median imputation).
- Converted all boolean columns (True/False) to integers (0/1).
- One-hot encoded categorical features (Contract, InternetService, PaymentMethod, etc.).
- Engineered features: Service_Intensity, Contract_Rank, Value_at_Risk (tenure $\times$ MonthlyCharges).
- Removed customerID and separated features from the target (Churn).

The final model-ready dataset contained 32 columns and was stratified/split (80/20) to preserve the ~26.5% churn rate.

Appendix B: Feature Engineering and Model Architecture

Engineered features (Service_Intensity, Contract_Rank, Value_at_Risk) were created to capture business risk more directly than raw variables. The final XGBoost model used 200–300 trees with optimized hyperparameters (learning_rate=0.05, max_depth=4, etc.).

Appendix C: Technical Methodology: Random Forest & XGBoost

Random Forest provided a strong baseline with class-weight balancing. XGBoost improved performance through gradient boosting and its handling of class imbalance via scale_pos_weight. Both models are ensemble tree-based methods that excel at capturing non-linear interactions in tabular customer data.

Appendix D: Evaluation Metrics

In retention modeling, missing a high-risk customer (False Negative) is far more costly than a False Positive. Therefore, we optimized for F1-score on the churn class and used AUC-ROC for overall ranking quality.

Figure 4: Confusion Matrix & Classification Report (Tuned XGBoost with Optimized Threshold)

```
Best threshold for F1-score: 0.5155
Best F1-score at that threshold: 0.6336

=== PERFORMANCE WITH OPTIMIZED THRESHOLD ===
      precision    recall  f1-score   support

     0       0.91      0.75      0.82      1035
     1       0.53      0.79      0.63       374

 accuracy      0.76      1409
 macro avg      0.72      0.77      0.73      1409
 weighted avg      0.81      0.76      0.77      1409

AUC-ROC: 0.8442
```

Figure 4